

FL-04: Automated Evaluation & Verification Loops

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1. System Overview & Verification Goals

Goal: Replaced manual data inspection with deterministic python assertions (`scripts/verify_data_contract.py`) and JSON receipt logging (`outputs/verification_receipt.json`).

2. Automated Test Suite Architecture

Executed Assertions (4/4 PASS):

- Check 1: File existence & row count check (`data/raw/content_refresh_anonymized.csv`, 30,000 rows).
- Check 2: Zero leakage guard (assert no forbidden features like `trend_direction` used in model matrix).
- Check 3: Schema & null value bounds (assert 0 nulls in core features).
- Check 4: Base rate lift validation (assert `LogReg Precision@50 0.5000 > base rate 0.2890`).

3. Persistent Receipt Artifact (`outputs/verification_receipt.json`)

```
{ "status": "PASS", "checks_passed": 4, "checks_failed": 0, "details": [ {"check": "Starter Dataset File Exists",
"status": "PASS", "rows": 30000}, {"check": "Zero Data Leakage Guard", "status": "PASS", "forbidden_used": 0},
{"check": "Schema & Null Bounds", "status": "PASS", "null_counts": {"impressions_90d": 0}}, {"check": "Model
Precision@50 Lift Over Base Rate", "status": "PASS", "precision_at_50": 0.5} ] }
```

4. Manual vs Automated Efficiency Comparison Table

Dimension	Manual Human Verification	Automated Assertion Loop
Execution Speed	~15–20 minutes per release	< 1.2 seconds
Leakage Detection	Prone to human oversight on diffs	100% deterministic assertion
CI Integration	Cannot block automated PRs	Blocks invalid commits (exit code 1)
Audit Trail	Informal manual notes	JSON receipts (<code>outputs/*.json</code>)

5. Key Reusable Verification Guidelines

- Deterministic Assertions:** Use binary code assertions for zero-tolerance safety rules.
- Persistent JSON Receipts:** Export JSON files to `outputs/` for CI reviewers.
- Exit Code 1 Halts:** Force python scripts to return exit code 1 to halt failing CI pipelines.